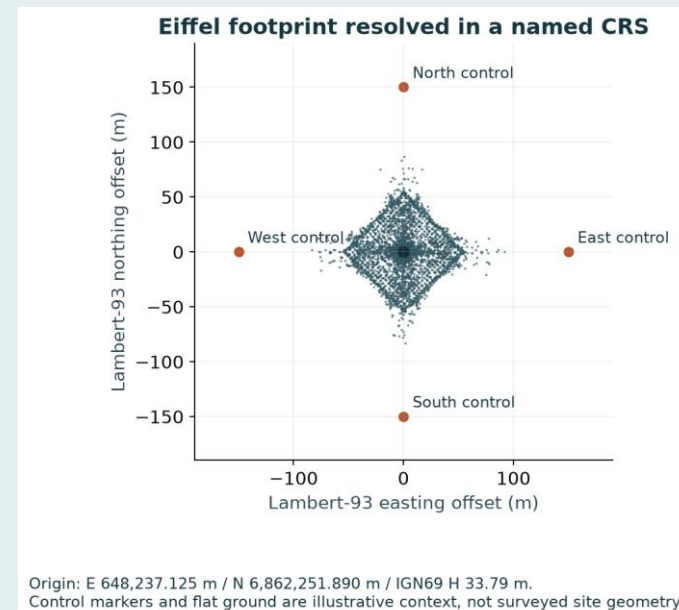
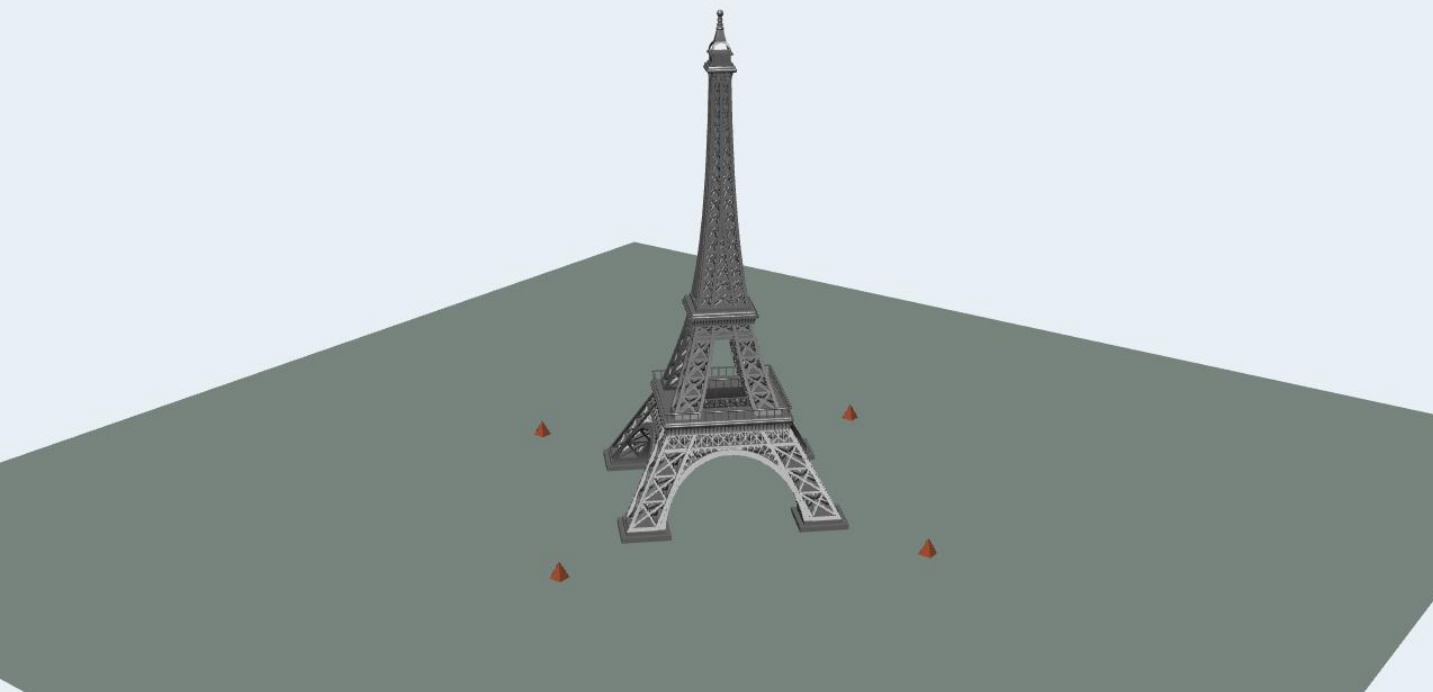


# Geospatial meaning that survives USD composition



The Eiffel model is placed at Lambert-93 E 648,237.125 m, N 6,862,251.890 m and IGN69 height 33.79 m, with a 45° CRS orientation. The same authored model resolves into Lambert-93, two UTM zones and ECEF.

Native Hydra render + named coordinate query.  
Illustrative controls; no survey accuracy claim.

# Numbers alone do not place an asset

A USD transform says how local geometry moves within a scene. It does not say which Earth reference defines a coordinate, which vertical datum defines its height, or how a site's calibrated grid relates to another project's coordinates.

## LOCAL USD

Point (3,4,5)  
Child translate (1,0,0)  
Model translate (7,8,9)

## ORDINARY READER

**(11,12,14)**

Ordinary transforms retain  
their ordinary interpretation.

## CRS-AWARE READER

**(1003,2016,44)**

Explicit position + orientation  
+ scale; ordinary transforms  
after.

A hand-worked noncommuting control catches an origin-only or wrong-order implementation.

# CRS meaning first. Ordinary USD transforms after.

A complete WKT definition identifies the coordinate reference system. A binding identifies which definition applies. Separate position, orientation and scale place a model; ordinary USD transforms apply after geospatial placement and resolution.

01

## COMPLETE WKT

Definition + component units  
Datum, height and site calibration

02

## BINDING

Source association  
Independent assets retain meaning

03

## CRS PLACEMENT

Position + orientation + scale  
Local geometry stays  
USD lengths

04

## ORDINARY USD

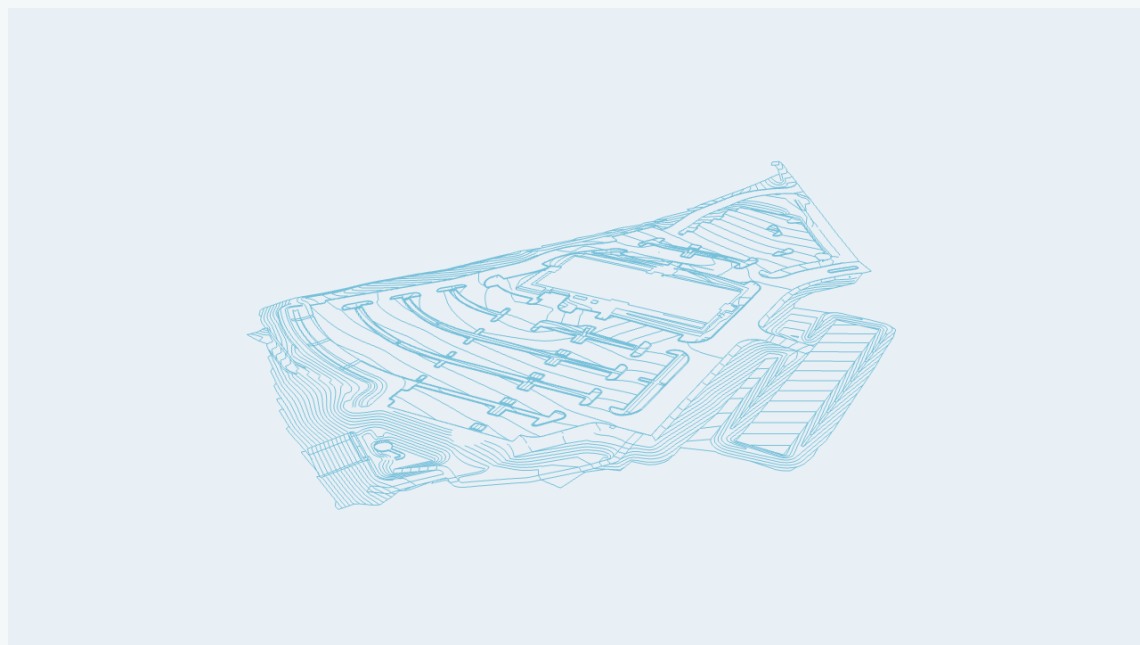
Post-placement adjustments  
Writer conformance stays separate

# Author meaning once. Derive results without altering source layers.

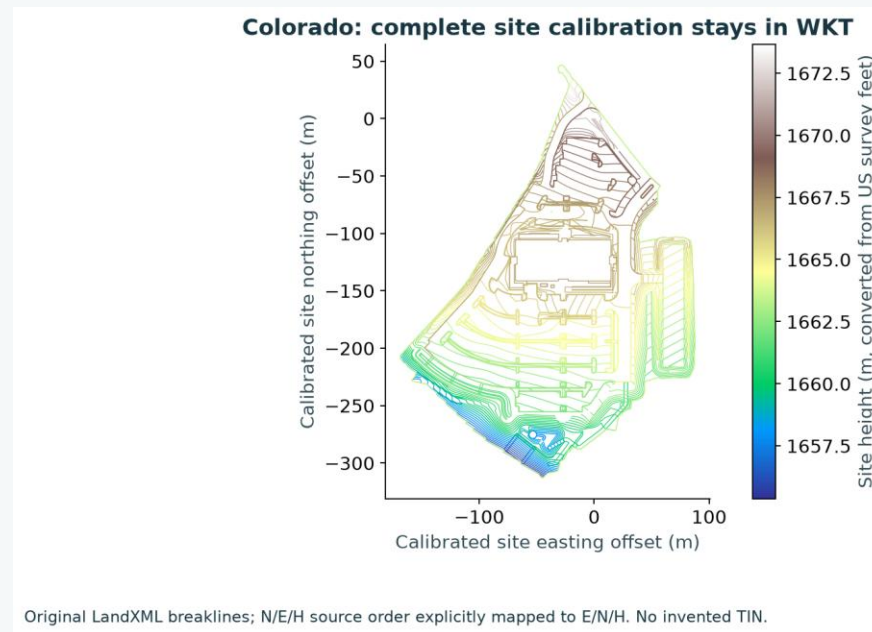
Exact experimental fields and adjustment context are frozen before code and remain reviewable choices.

# Complete CRS definitions do real work

The partner data exercise geographic, projected, compound height and derived site-calibration definitions. Colorado keeps US survey feet and its full horizontal/vertical site calibration in WKT; France keeps Lambert-93/CC49 and IGN69 height meaning.



**Native Hydra / original LandXML**



**Headless query / calibrated site axes**

5 Colorado + 59 France controls • Required height grids verified • CSVs use the same PROJ family

# The same authored data reaches three consumers

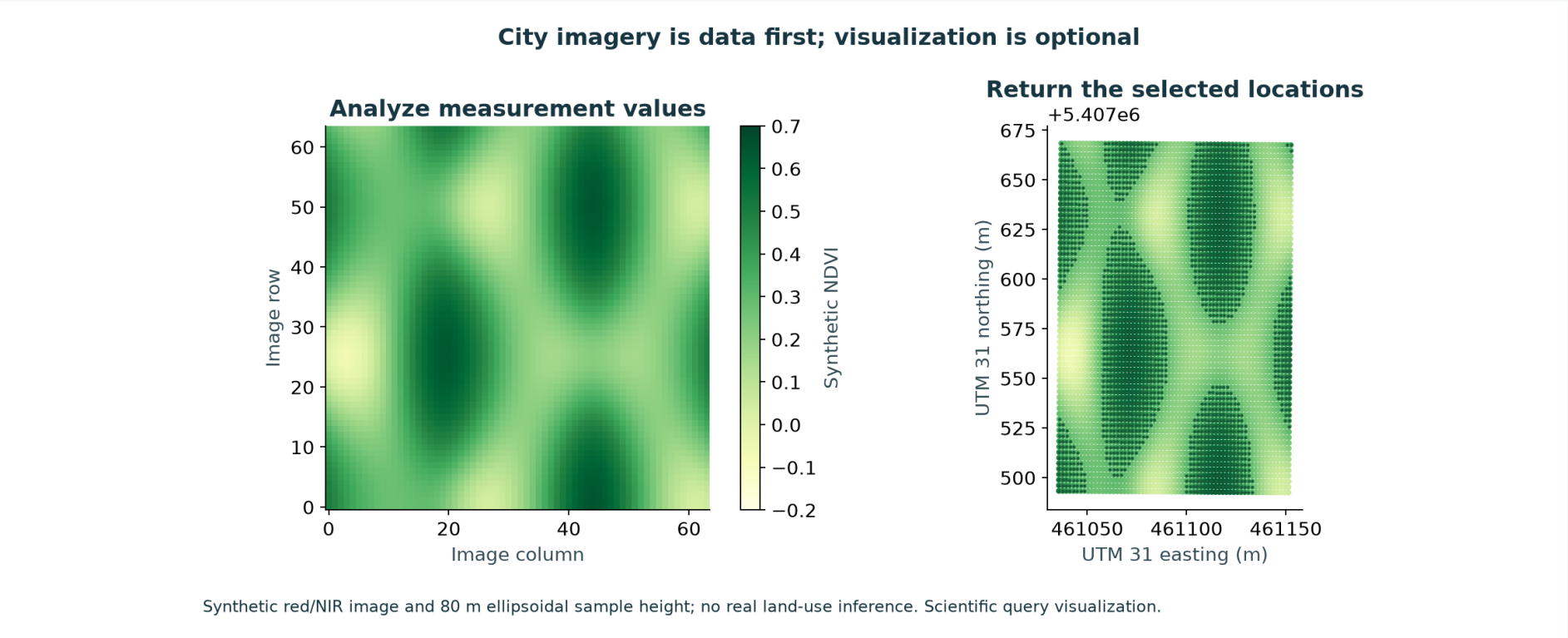
Headless USD queries, the native C++ Hydra resolver and the separate live OV resolver agree across all 27 jobs. Each derives placements from the authored scene; neither runtime consumes a headless result snapshot.

| Dataset                  | Jobs | Native max (mm) | OV max (mm) |
|--------------------------|------|-----------------|-------------|
| Eiffel model + context   | 4    | 0.000000        | 0.000000    |
| Colorado breaklines      | 3    | 0.000104        | 0.000000    |
| Original railway         | 3    | 0.000002        | 0.000000    |
| Synthetic city imagery   | 2    | 0.000000        | 0.000000    |
| Global scalar grid       | 3    | 0.000000        | 0.000000    |
| Composition + time       | 3    | 0.000002        | 0.000000    |
| Native + point instances | 2    | 0.000000        | 0.000000    |
| Partner controls         | 7    | 0.000000        | 0.000000    |

Actual Hydra matrices / instancers + live OV result readback. Shared PROJ: numerical agreement ≠ independent geodetic accuracy.

# City analysis returns data, not just a picture

A 64 × 64 synthetic red/NIR image yields 1,850 samples above NDVI 0.3. The result returns locations in the requested CRS while preserving source indices, values and observation times.



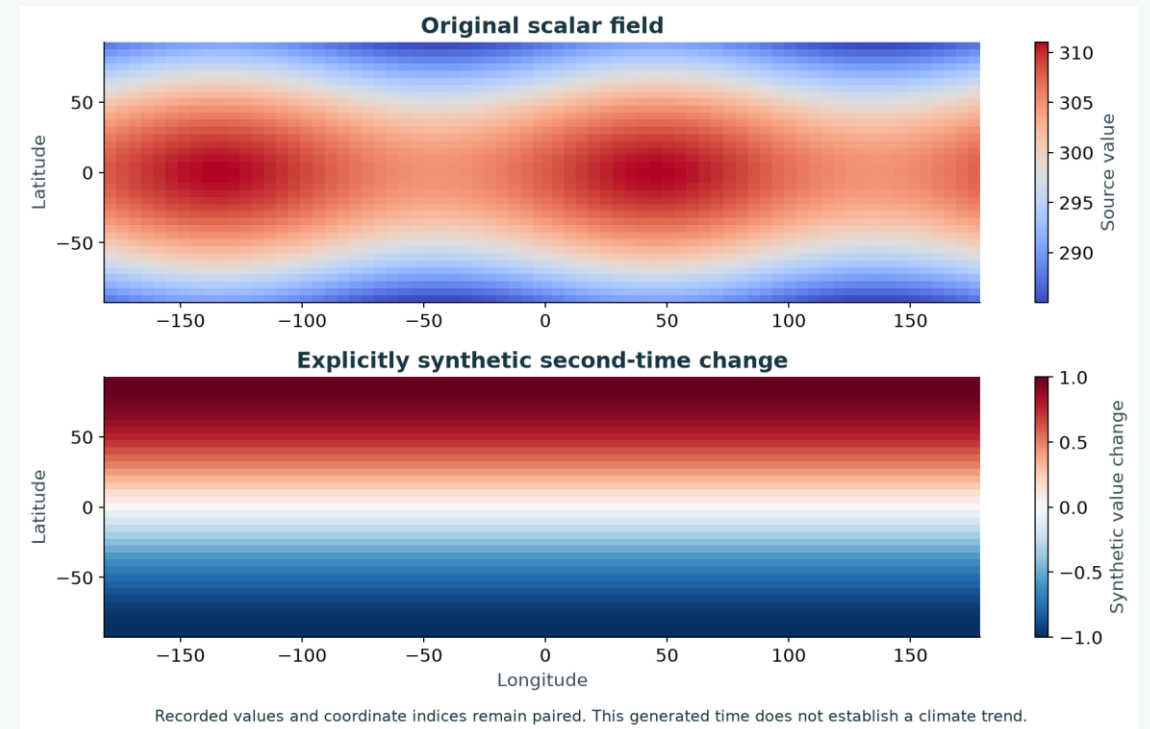
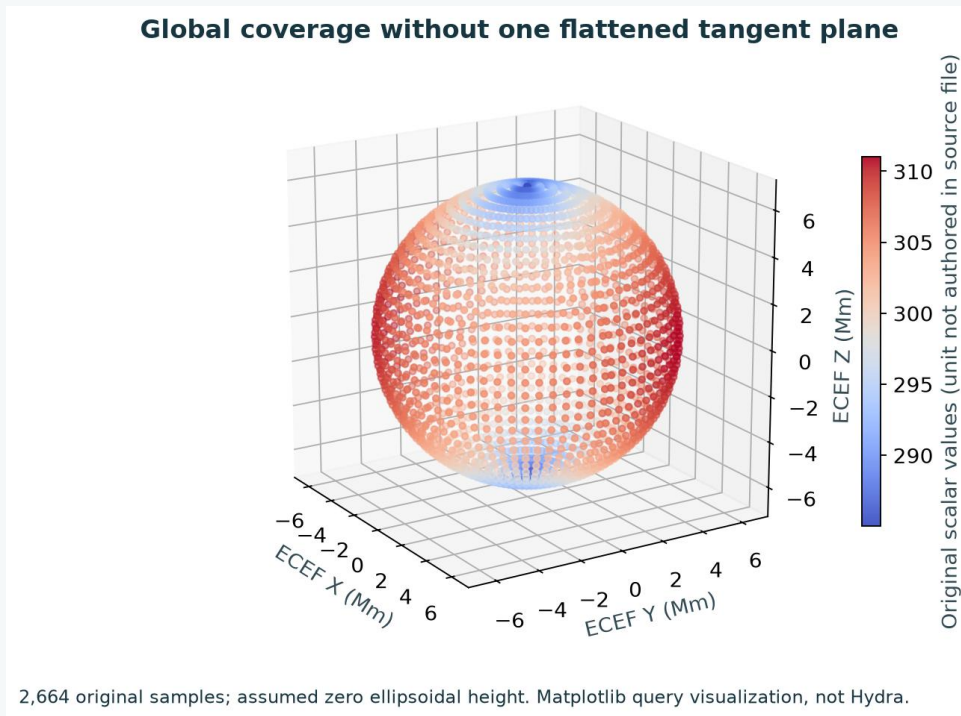
1,850  
selected  
locations

Coordinates  
Values  
Source indices  
Recorded  
times

Synthetic red/NIR and assumed 80 m ellipsoidal height. GeoJSON + paired analysis product accompany the run.

# Global coverage with coordinates and values intact

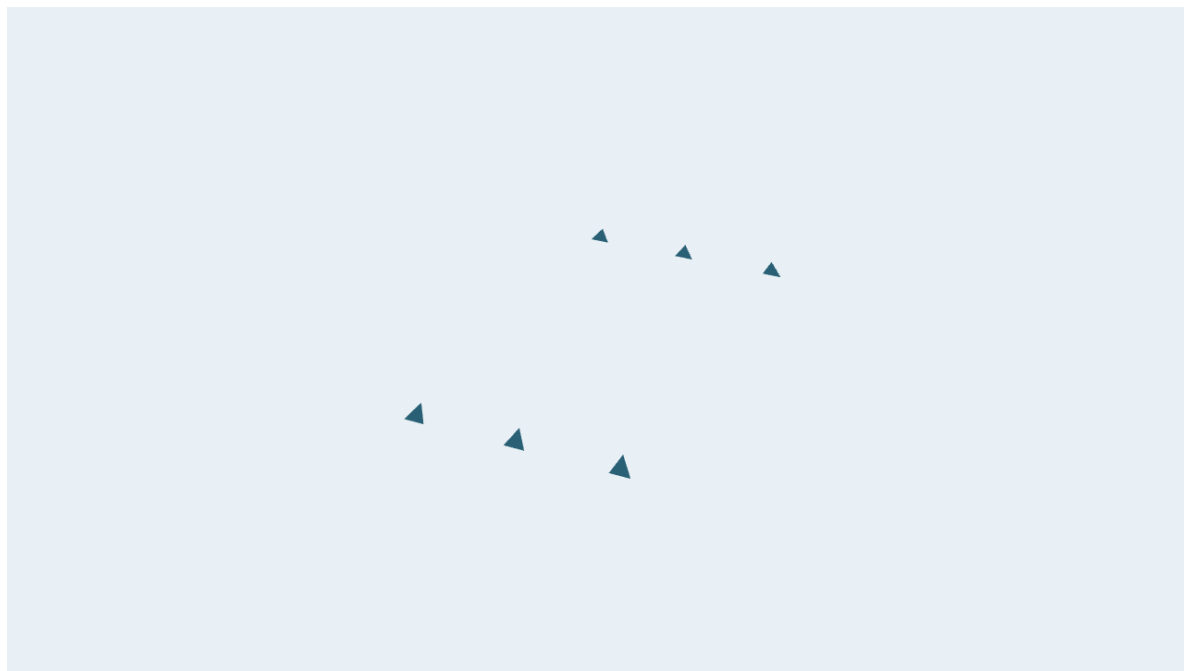
The original  $37 \times 72$  scalar grid resolves all 2,664 locations globally in ECEF and geographic coordinates. Values remain associated with their coordinates and grid indices across two recorded times.



Scientific query visualizations • Original scalar values / units absent in source • Second time explicitly synthetic

# Composition and live edits reach real consumers

Three USD native instances and three point instances share geometry while receiving distinct resolved placements. References, stronger opinions, variants, source interpolation, stage units and Y/Z up-axis are exercised with independent controls.



## Source edit → consumed placement

Native source notices recompute rendered placement; +10 m edit verified.

## Failure → stale results removed → recovery

Live OV hides invalid model results, clears measurement results, then recovers.

Five explicit exports re-resolved by a fresh reader. Values and recorded times preserved; no private skip flag.

# Precision is measured. Extent needs a contract.

Double coordinate anchors and small local floats preserve detail away from the origin. But one affine model frame approximates a nonlinear transformation, and the error grows with the model's spatial extent.

| Half-extent | Points | Affine discrepancy |
|-------------|--------|--------------------|
| ±100 m      | 289    | 1.57 mm            |
| ±1,000 m    | 289    | 0.157 m            |
| ±10,000 m   | 289    | 15.7 m             |

## SAMPLED DISCREPANCY

Finite samples do not establish a continuous surface error bound.

Double anchor + small local floats avoids 31.25 mm float spacing at a 481,948 m absolute coordinate.

R24 needs an extent / approximation contract. This is a design finding, not a deferred demonstration.

# Executed evidence separates choices from gaps

The executed candidate provides reviewable answers for exact placement fields, adjustment context, geographic/geocentric tuples, WKT string normalization, measurement association, dependency declarations and sampled export records. The group still needs to choose those contracts and settle the continuous extent/error requirement.

## REVIEW THE EXPERIMENTAL CONTRACTS

- Exact placement fields and axis conventions
- Project adjustment context
- WKT string normalization
- Measurement association
- Dependency and sampled-export records

## PRESERVE THE ROADMAP

- Coordinate epochs fully deferred.
- CRS frame epochs retained in WKT.
- Supported datum / height operations exercised.
- City and global data remain in scope.
- Continuous extent / error bound stays open.

Requirements → frozen model + runtime → independent consumers → audit → evidence → delivery